UNIVERSITY OF LEEDS

Relational Data

Deep Learning - COMP5203M

2022-23

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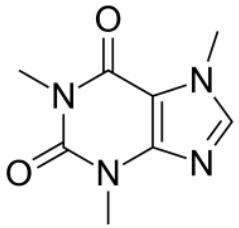
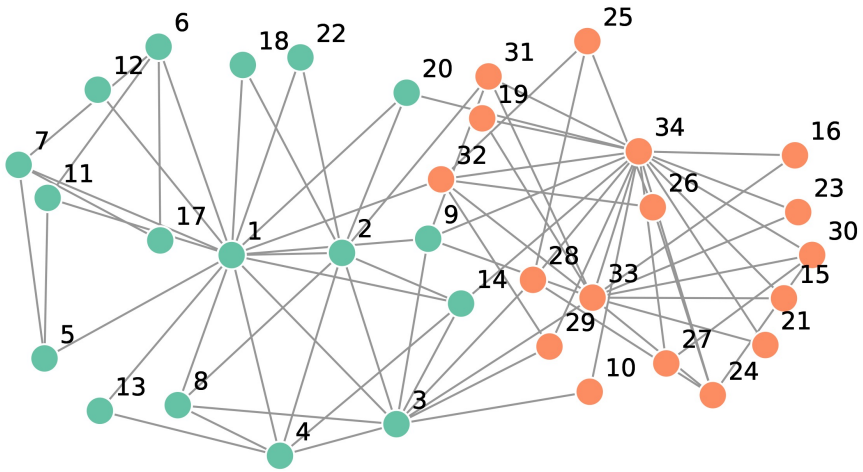
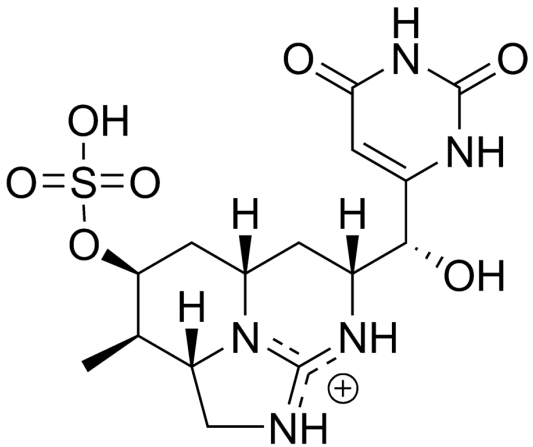
Graphs

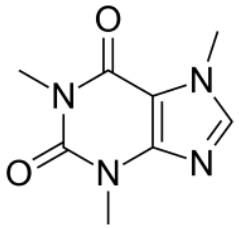




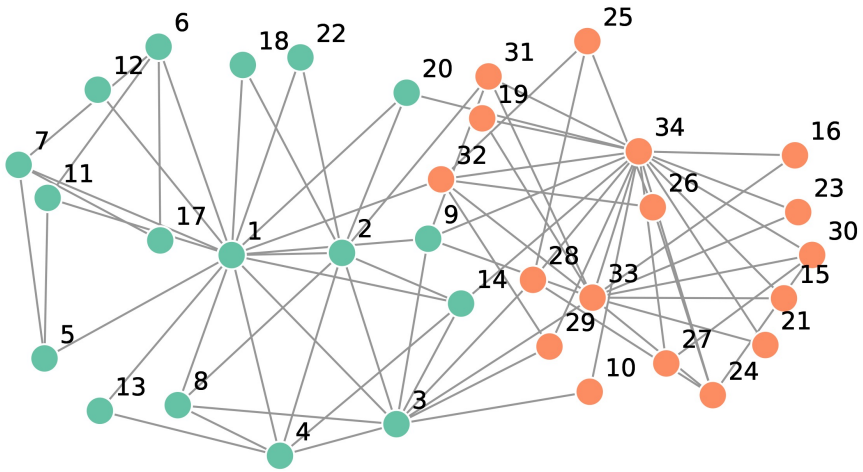


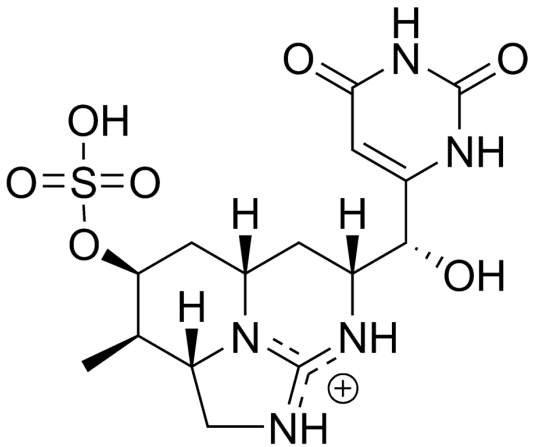
https://en.wikipedia.org/wiki/Cyanotoxin
https://en.wikipedia.org/wiki/Molecular_graph
https://en.wikipedia.org/wiki/Zachary%27s_karate_club

Graphs



Pungent?





Toxic?

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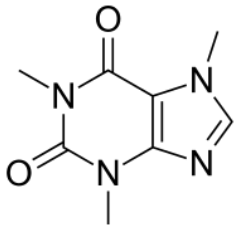
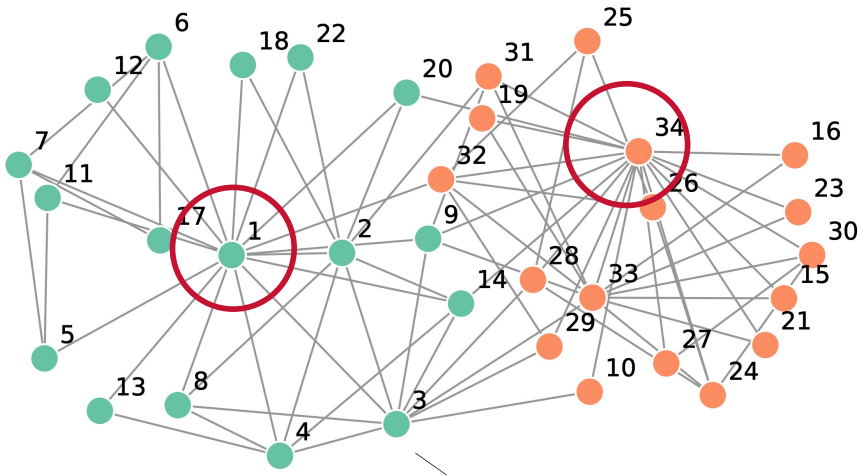
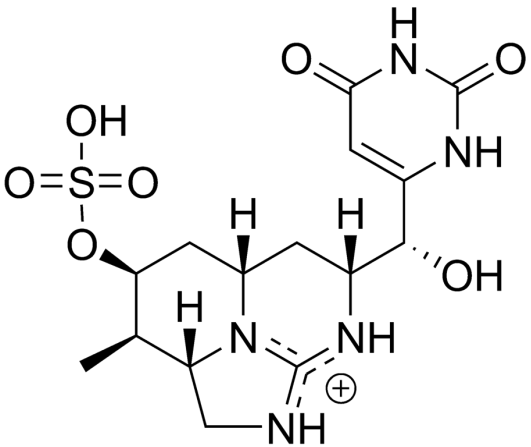
Graphs



Pungent?



Who joins each influential person?



Toxic?

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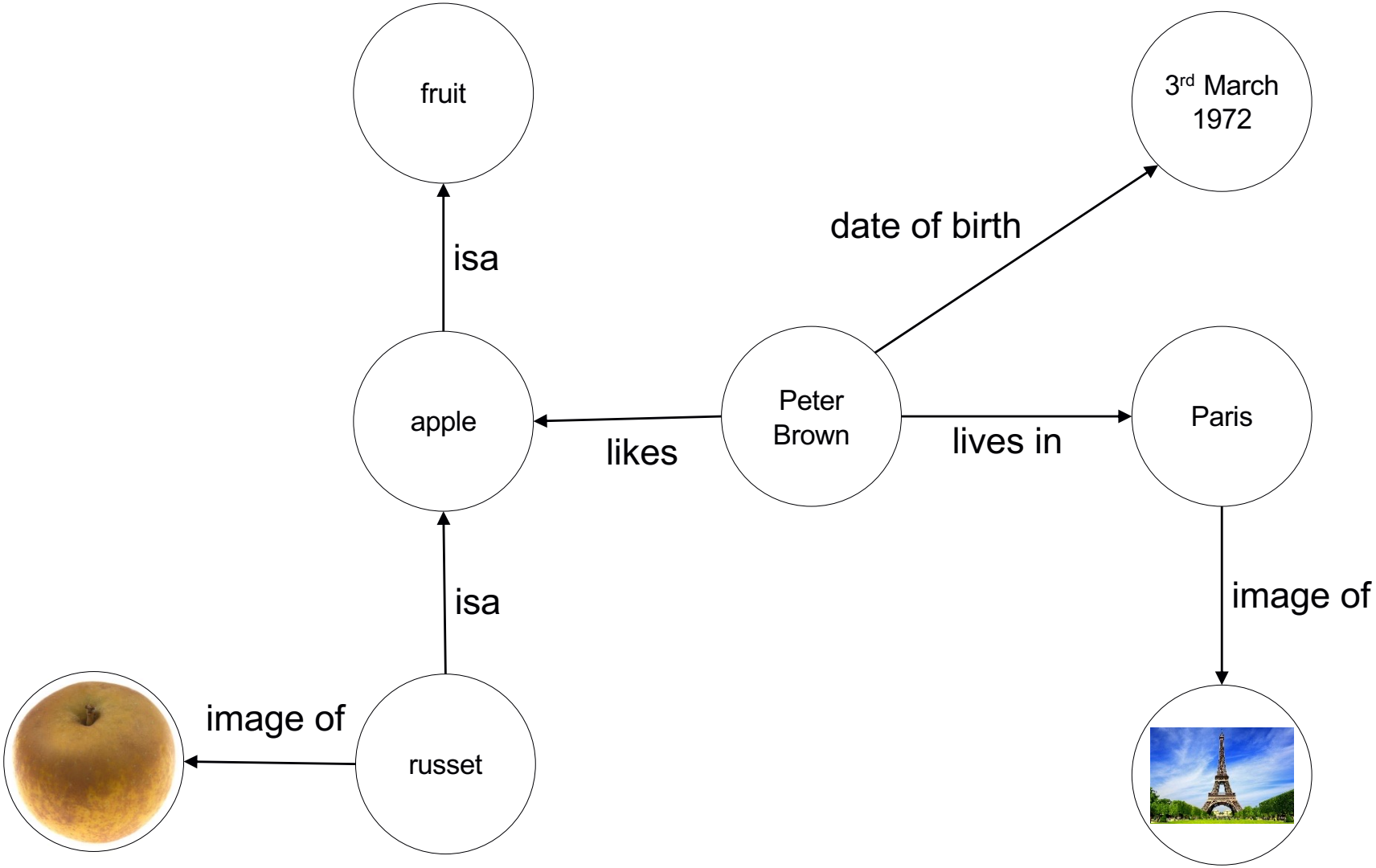
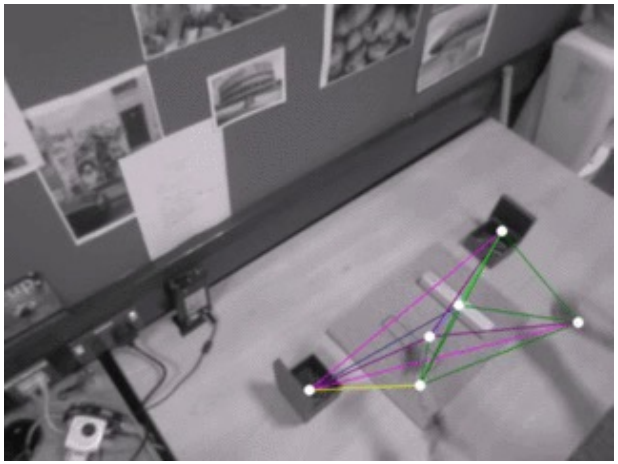
Knowledge Graphs



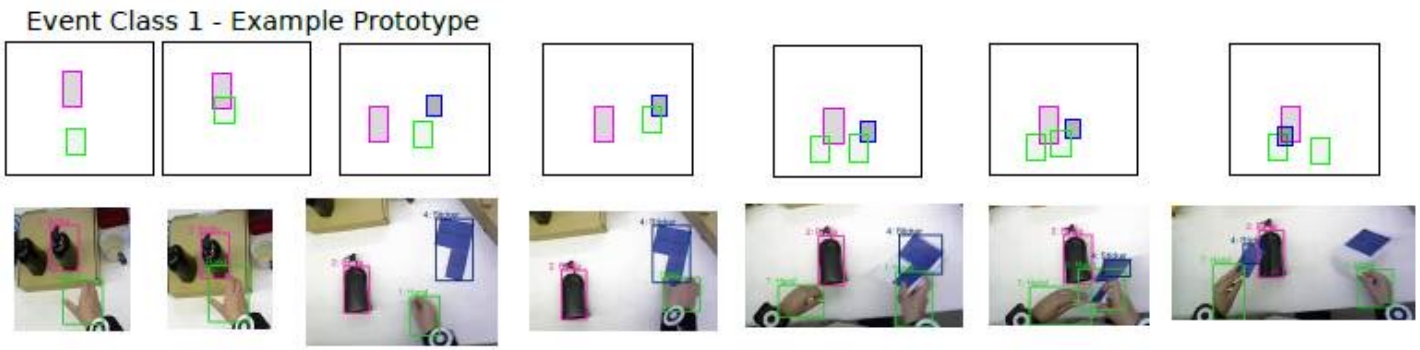
Typical tasks:

- Inferring new information (e.g. Peter Brown likes russet apples)
- Adding to incomplete information

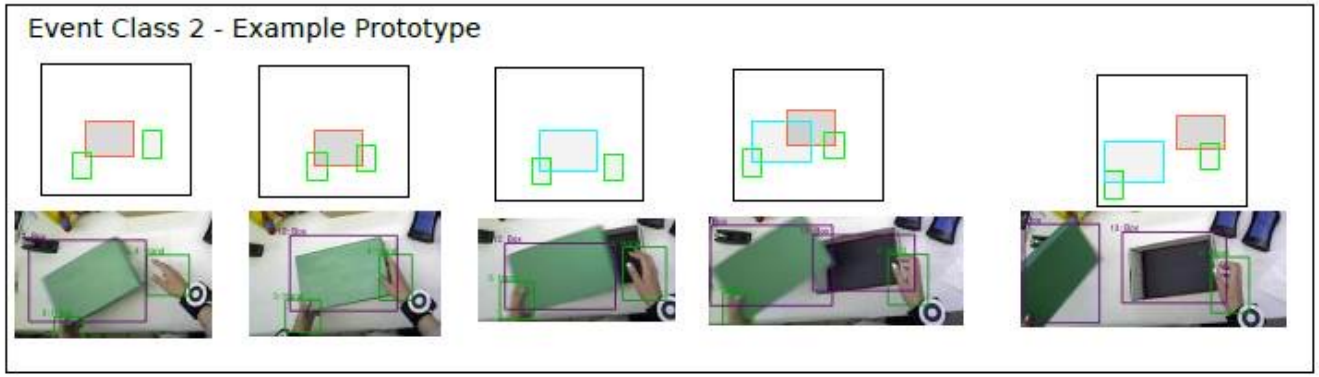
Learning about activities from the changing spatial relations between objects



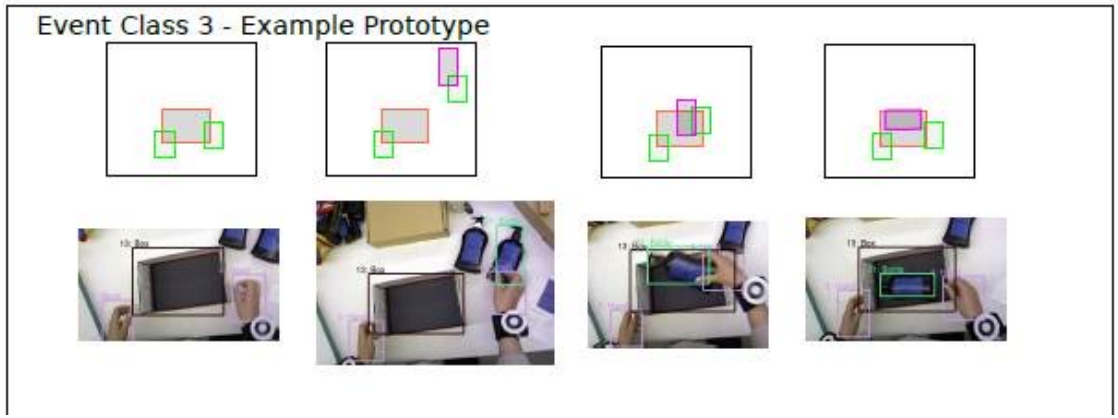
Emergent event classes



“putting the sticker on the bottle”

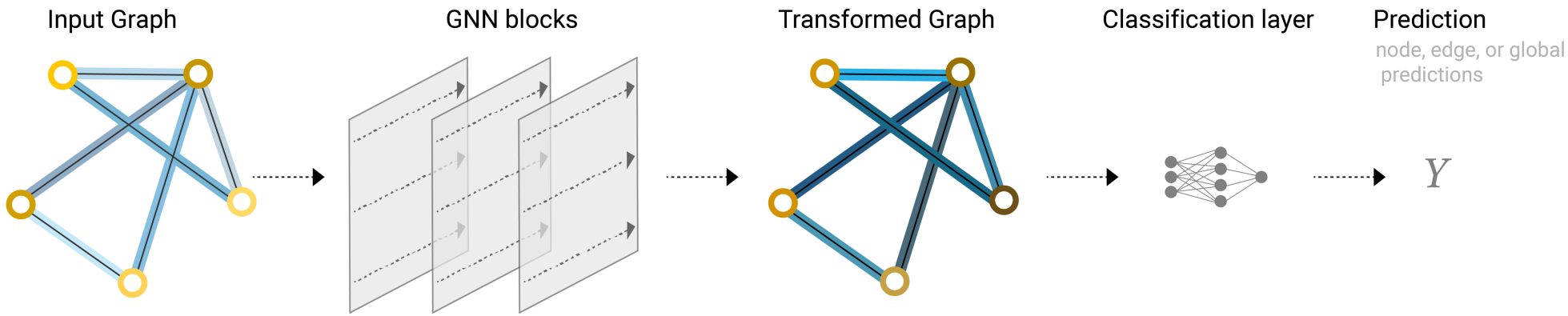


“opening the cover of the box”



“putting the bottle inside the box”

Graph neural networks



https://distill.pub/2021/gnn-intro/

Inductive bias

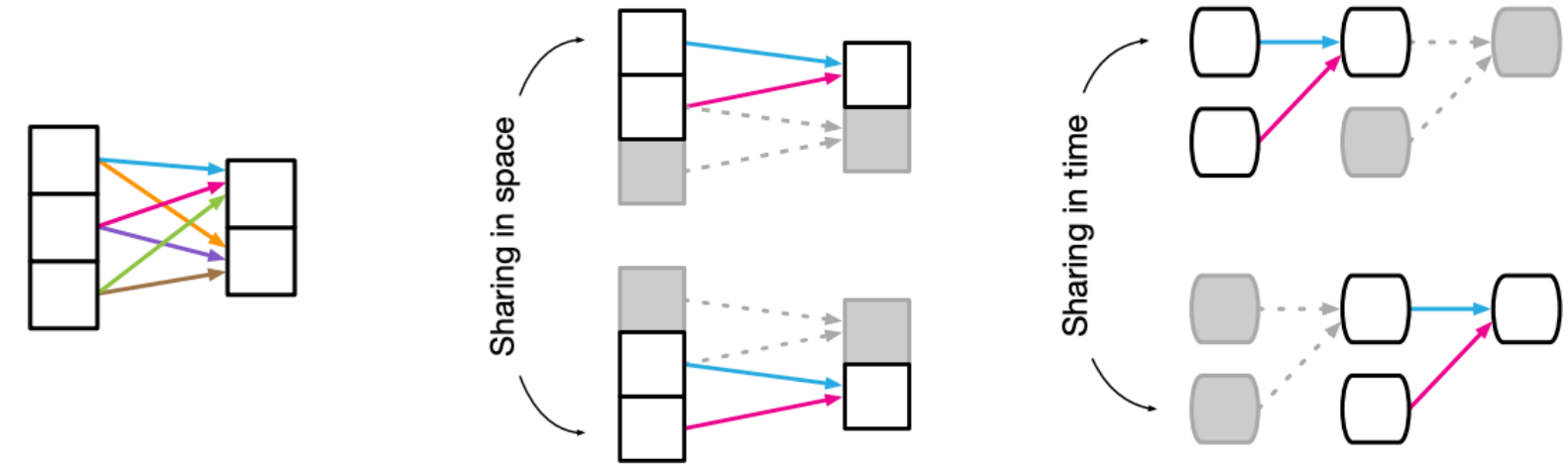
"An inductive bias allows a learning algorithm to prioritize one solution (or interpretation) over another, independent of the observed data (Mitchell, 1980)."

Examples:

- Prior distribution in Bayesian probability.
- Regularisation.
- Architecture of neural networks.

Battaglia, Peter W., et al. "Relational inductive biases, deep learning, and graph networks." *arXiv preprint arXiv:1806.01261* (2018).

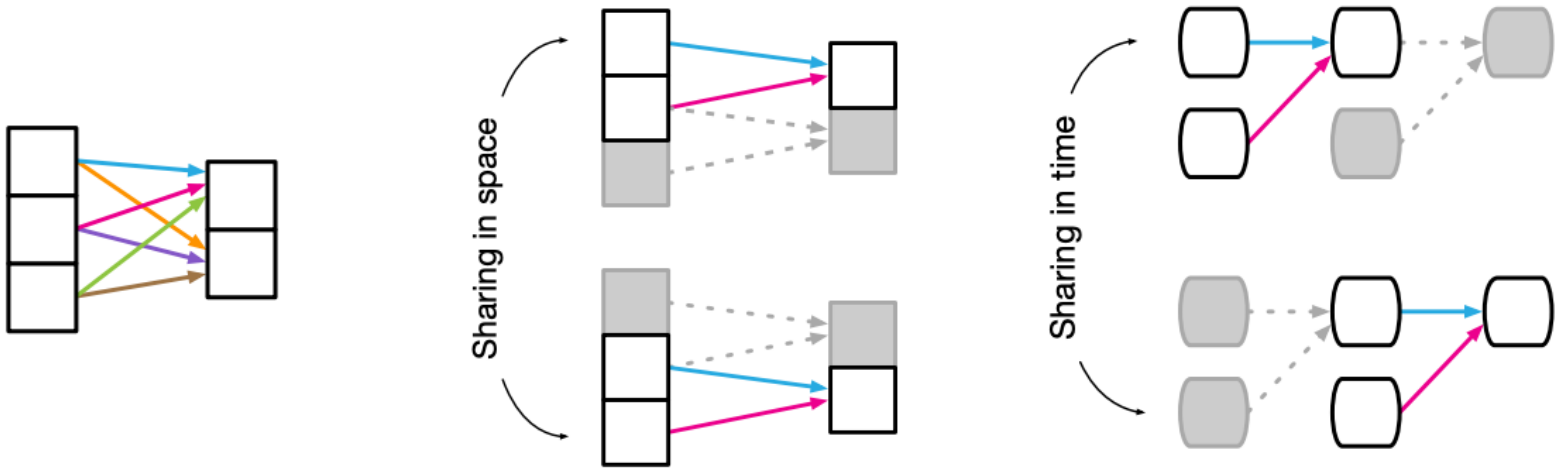
Inductive bias



(a) Fully connected (b) Convolutional (c) Recurrent

Battaglia, Peter W., et al. "Relational inductive biases, deep learning, and graph networks." *arXiv preprint arXiv:1806.01261* (2018).

Inductive bias

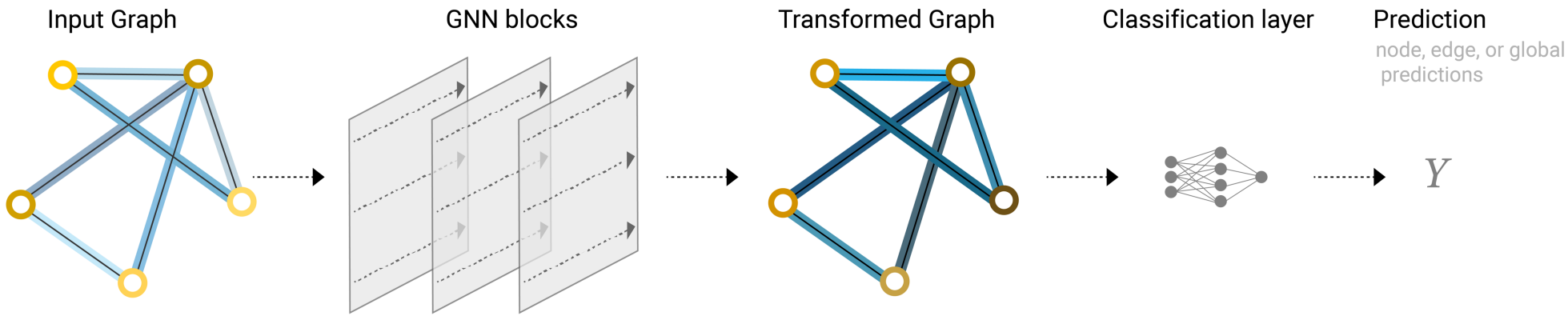


(a) Fully connected (b) Convolutional (c) Recurrent

Component	Entities	Relations	Rel. inductive bias	Invariance
Fully connected	Units	All-to-all	Weak	-
Convolutional	Grid elements	Local	Locality	Spatial translation
Recurrent	Timesteps	Sequential	Sequentiality	Time translation
Graph network	Nodes	Edges	Arbitrary	Node, edge permutations

Battaglia, Peter W., et al. "Relational inductive biases, deep learning, and graph networks." *arXiv preprint arXiv:1806.01261* (2018).

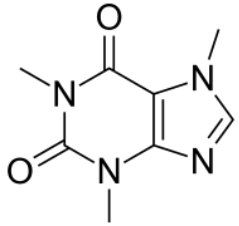
Graph neural networks

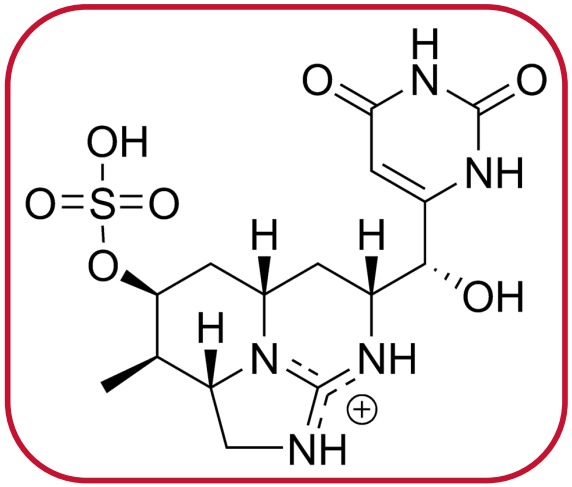


https://distill.pub/2021/gnn-intro/

Various prediction tasks

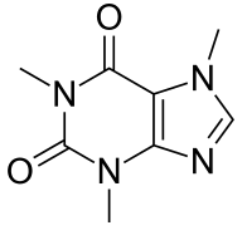
- Graph-level tasks

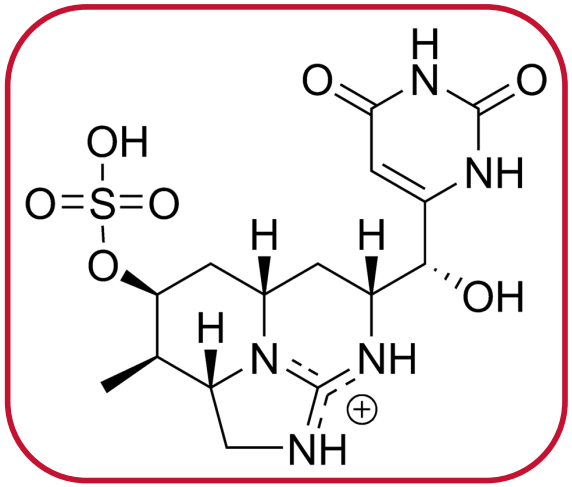




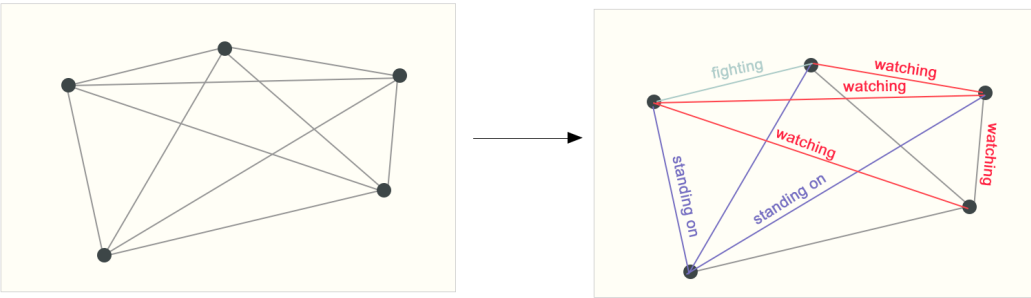
Various prediction tasks

- Graph-level tasks



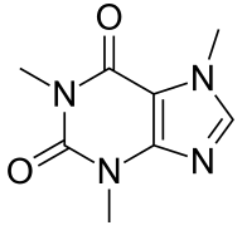


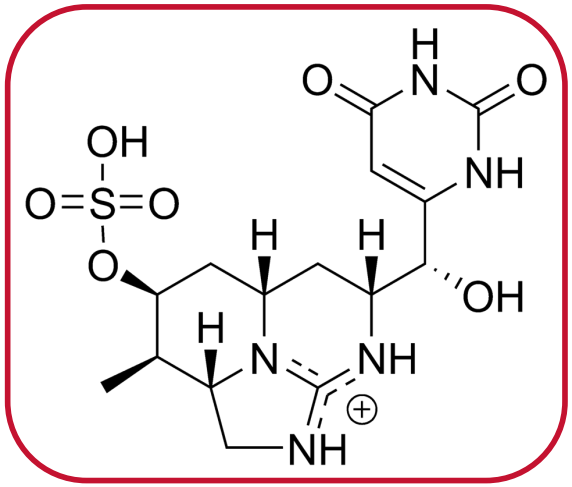
- Edge-level tasks



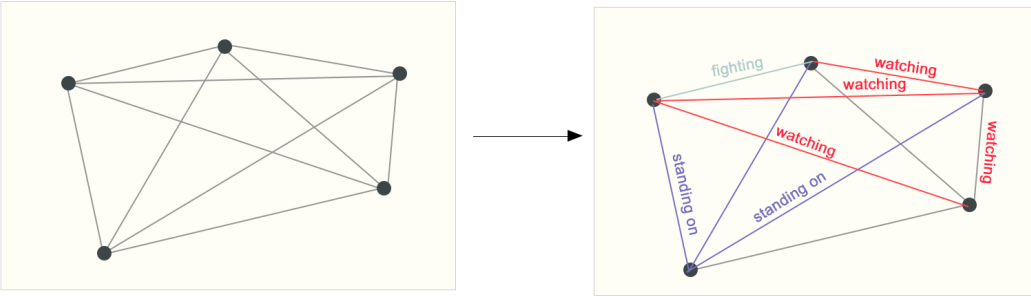
Various prediction tasks

- Graph-level tasks

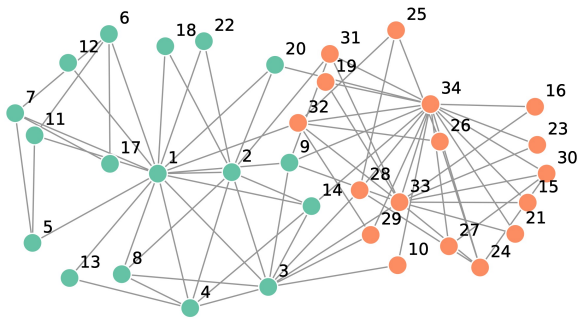




- Edge-level tasks



- Node-level tasks



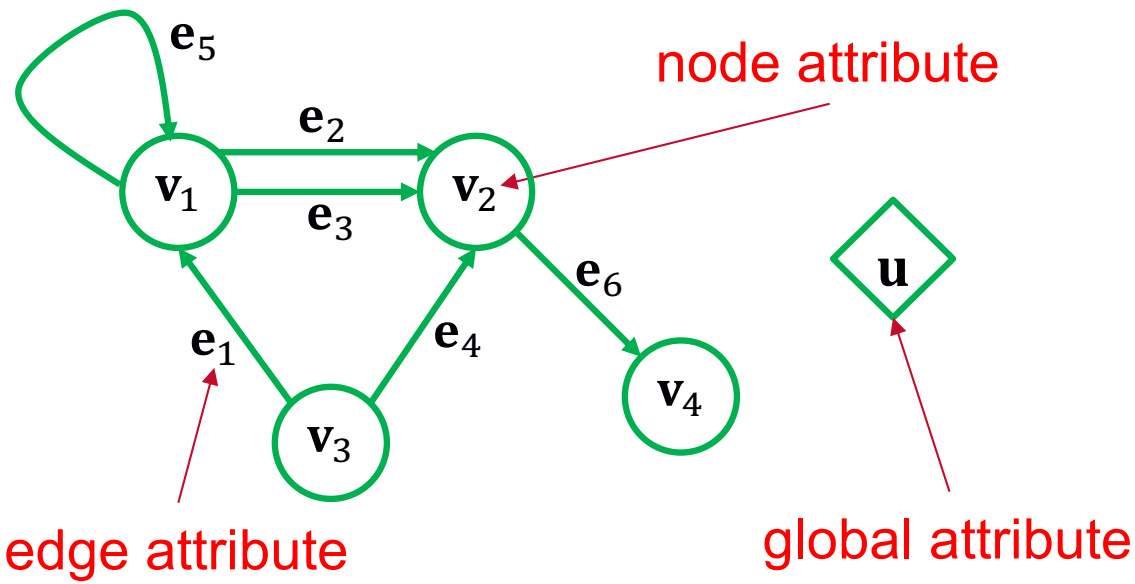
Defining a *graph* in the GN framework

$$G = (\mathbf{u}, V, E)$$

$$V = \{(\mathbf{v}_i, i)\}_{i=1:N^v}$$

$$E = \{(\mathbf{e}_k, r_k, s_k)\}_{k=1:N^e}$$

A (GN) *graph* is made up of N^v nodes (vertices) V, N^e edges E connecting the nodes, and a global attribute $\mathbf{u}$.



$$V = \{(\mathbf{v}_1, 1), (\mathbf{v}_2, 2), (\mathbf{v}_3, 3), (\mathbf{v}_4, 4)\}$$

$$E = \{(\mathbf{e}_1, 1{,}3), (\mathbf{e}_2, 2{,}1), (\mathbf{e}_3, 2{,}1), (\mathbf{e}_4, 2{,}3), (\mathbf{e}_5, 1{,}1), (\mathbf{e}_6, 4{,}2)\}$$

Node, edge and global attributes

Node, edge and global attributes are tensors (e.g. word embeddings, image patches).

nodes and edges have attributes

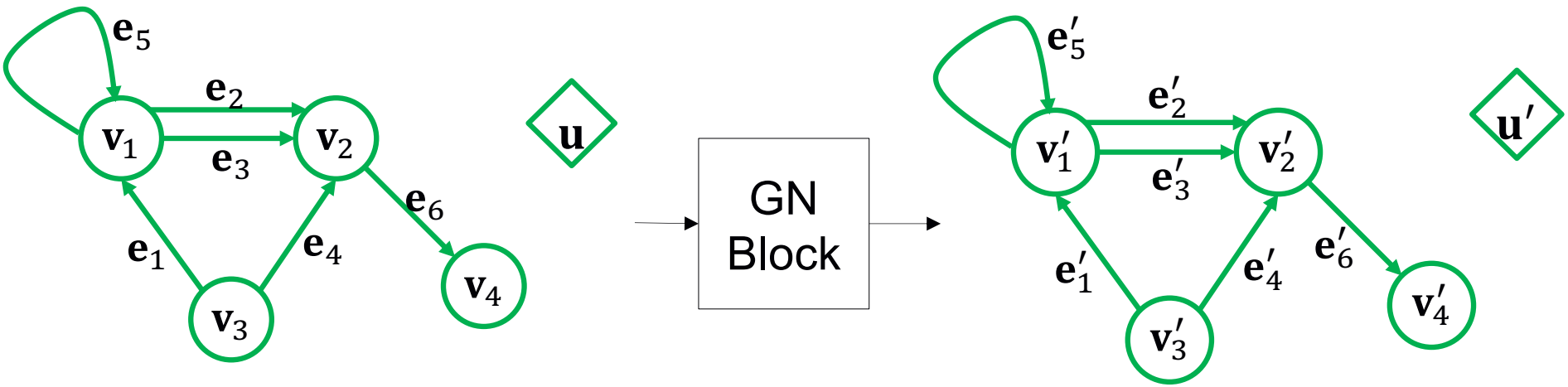
In standard terms, this is a directed, attributed multi-graph with a global attribute

one-way edges

may be more than one edge between nodes

GN Block

A GN block transforms a (GN) graph by updating the edge, node and global attributes. It takes a graph as input and outputs a modified graph.



The transformation involves six steps.

GN Block

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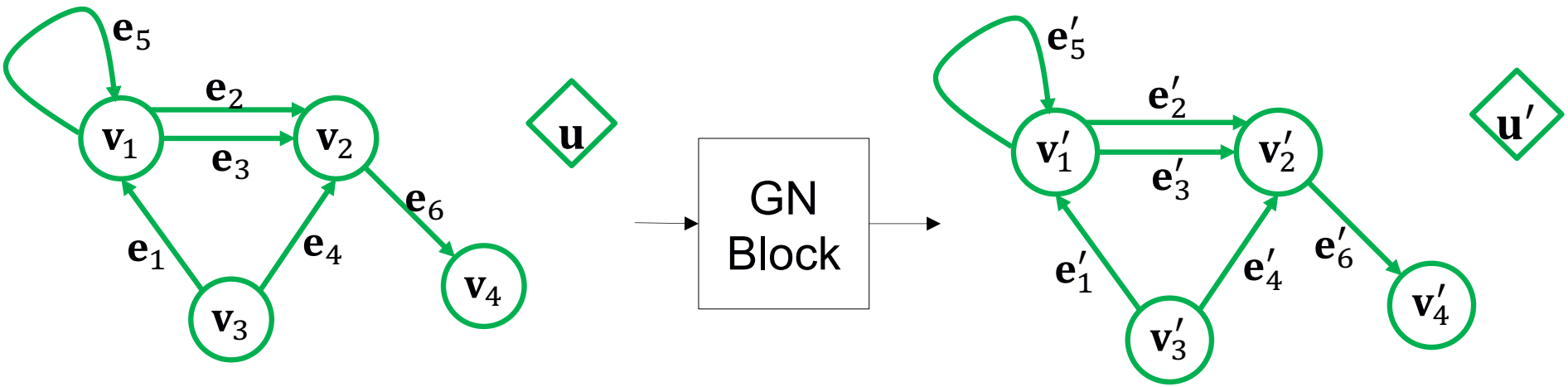

$$\mathbf{e}'_k = \phi^e\left(\mathbf{e}_k, \mathbf{v}_{r_k}, \mathbf{v}_{s_k}, \mathbf{u}\right)$$
$$\mathbf{v}'_i = \phi^v\left(\bar{\mathbf{e}}'_i, \mathbf{v}_i, \mathbf{u}\right)$$
$$\mathbf{u}' = \phi^u\left(\bar{\mathbf{e}}', \bar{\mathbf{v}}', \mathbf{u}\right)$$

$$\bar{\mathbf{e}}'_i = \rho^{e\to v}\left(E'_i\right)$$
$$\bar{\mathbf{e}}' = \rho^{e\to u}\left(E'\right)$$
$$\bar{\mathbf{v}}' = \rho^{v\to u}\left(V'\right)$$

GNN Block Algorithm

Algorithm 1 Steps of computation in a full GN block.

```
function GraphNetwork(E, V, u)
    for k ∈ {1 ... N^e} do
        e'_k ← φ^e (e_k, v_{r_k}, v_{s_k}, u)          ▷ 1. Compute updated edge attributes
    end for
    for i ∈ {1 ... N^n} do
        let E'_i = {(e'_k, r_k, s_k)}_{r_k=i, k=1:N^e}
        ē'_i ← ρ^{e→v} (E'_i)                          ▷ 2. Aggregate edge attributes per node
        v'_i ← φ^v (ē'_i, v_i, u)                      ▷ 3. Compute updated node attributes
    end for
    let V' = {v'}_{i=1:N^v}
    let E' = {(e'_k, r_k, s_k)}_{k=1:N^e}
    ē' ← ρ^{e→u} (E')                                  ▷ 4. Aggregate edge attributes globally
    v̄' ← ρ^{v→u} (V')                                  ▷ 5. Aggregate node attributes globally
    u' ← φ^u (ē', v̄', u)                               ▷ 6. Compute updated global attribute
    return (E', V', u')
end function
```

GNN Block Algorithm

Algorithm 1 Steps of computation in a full GN block.

function GRAPHNETWORK(E, V, $\mathbf{u}$)

for $k \in \{1 \ldots N^e\}$ **do**

$\mathbf{e}'_k \leftarrow \phi^e(\mathbf{e}_k, \mathbf{v}_{r_k}, \mathbf{v}_{s_k}, \mathbf{u})$ ▷ 1. Compute updated edge attributes

end for

for $i \in \{1 \ldots N^n\}$ **do**

let $E'_i = \{(\mathbf{e}'_k, r_k, s_k)\}_{r_k=i,\ k=1:N^e}$

$\bar{\mathbf{e}}'_i \leftarrow \rho^{e \to v}(E'_i)$ ▷ 2. Aggregate edge attributes per node

$\mathbf{v}'_i \leftarrow \phi^v(\bar{\mathbf{e}}'_i, \mathbf{v}_i, \mathbf{u})$ ▷ 3. Compute updated node attributes

end for

let $V' = \{\mathbf{v}'\}_{i=1:N^v}$

let $E' = \{(\mathbf{e}'_k, r_k, s_k)\}_{k=1:N^e}$

$\bar{\mathbf{e}}' \leftarrow \rho^{e \to u}(E')$ ▷ 4. Aggregate edge attributes globally

$\bar{\mathbf{v}}' \leftarrow \rho^{v \to u}(V')$ ▷ 5. Aggregate node attributes globally

$\mathbf{u}' \leftarrow \phi^u(\bar{\mathbf{e}}', \bar{\mathbf{v}}', \mathbf{u})$ ▷ 6. Compute updated global attribute

return $(E', V', \mathbf{u}')$

end function

GNN Block Algorithm

Algorithm 1 Steps of computation in a full GN block.

function GRAPHNETWORK(E, V, $\mathbf{u}$)
  for $k \in \{1 \ldots N^e\}$ **do**
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  end for
  for $i \in \{1 \ldots N^n\}$ **do**
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    $\bar{\mathbf{e}}'_i \leftarrow \rho^{e\to v}\left(E'_i\right)$ ▷ 2. Aggregate edge attributes per node
    $\mathbf{v}'_i \leftarrow \phi^v\left(\bar{\mathbf{e}}'_i, \mathbf{v}_i, \mathbf{u}\right)$ ▷ 3. Compute updated node attributes
  end for
  let $V' = \left\{\mathbf{v}'\right\}_{i=1:N^v}$
  let $E' = \left\{\left(\mathbf{e}'_k, r_k, s_k\right)\right\}_{k=1:N^e}$
  $\bar{\mathbf{e}}' \leftarrow \rho^{e\to u}\left(E'\right)$ ▷ 4. Aggregate edge attributes globally
  $\bar{\mathbf{v}}' \leftarrow \rho^{v\to u}\left(V'\right)$ ▷ 5. Aggregate node attributes globally
  $\mathbf{u}' \leftarrow \phi^u\left(\bar{\mathbf{e}}', \bar{\mathbf{v}}', \mathbf{u}\right)$ ▷ 6. Compute updated global attribute
  return $\left(E', V', \mathbf{u}'\right)$
end function

GNN Block Algorithm

Algorithm 1 Steps of computation in a full GN block.

function GRAPHNETWORK(E, V, $\mathbf{u}$)
  for $k \in \{1 \ldots N^e\}$ **do**
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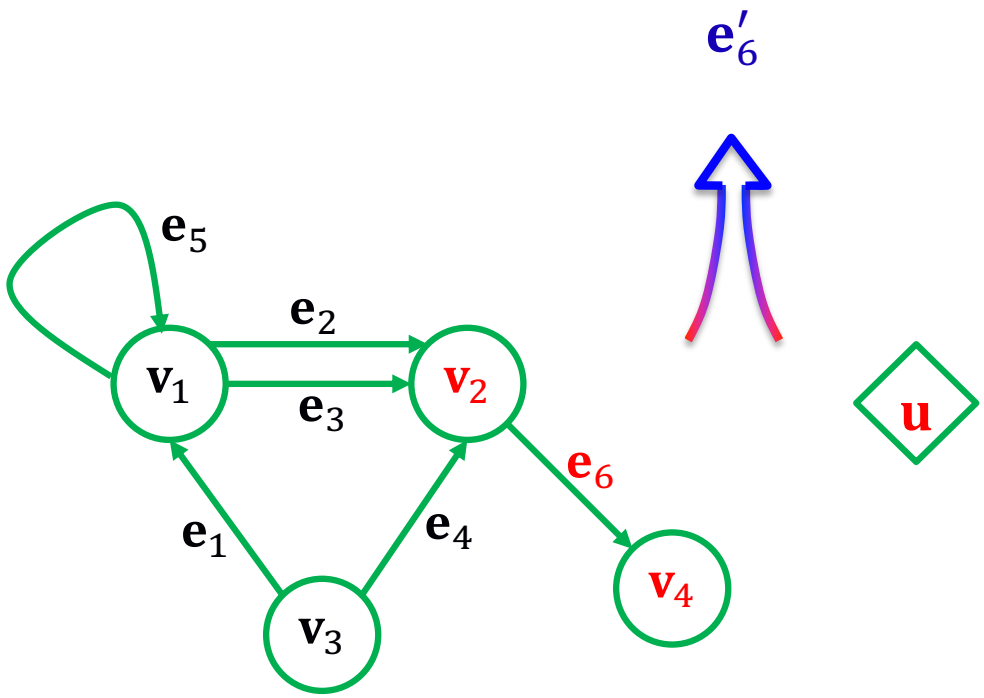
  $\mathbf{u}' \leftarrow \phi^u\left(\bar{\mathbf{e}}', \bar{\mathbf{v}}', \mathbf{u}\right)$ ▷ 6. Compute updated global attribute
  return $\left(E', V', \mathbf{u}'\right)$
end function

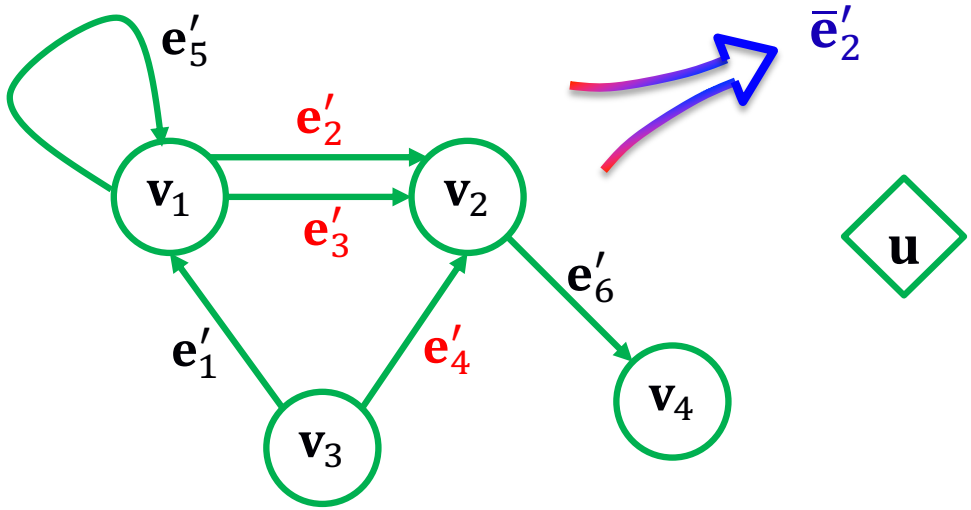
Edge update



Apply to all edge attributes (can be done in parallel)

$$\mathbf{e}'_k = \varphi^e\left(\mathbf{e}_k, \mathbf{v}_{r_k}, \mathbf{v}_{s_k}, \mathbf{u}\right)$$

Local aggregation of edge attributes



For each node, aggregate the attributes of incoming edges.

The aggregation function must be invariant to permutations of the inputs.

Typically use sum, mean, or maximum. For example, using mean:

$$\bar{\mathbf{e}}'_2 = (\mathbf{e}'_2 + \mathbf{e}'_3 + \mathbf{e}'_4)/3$$

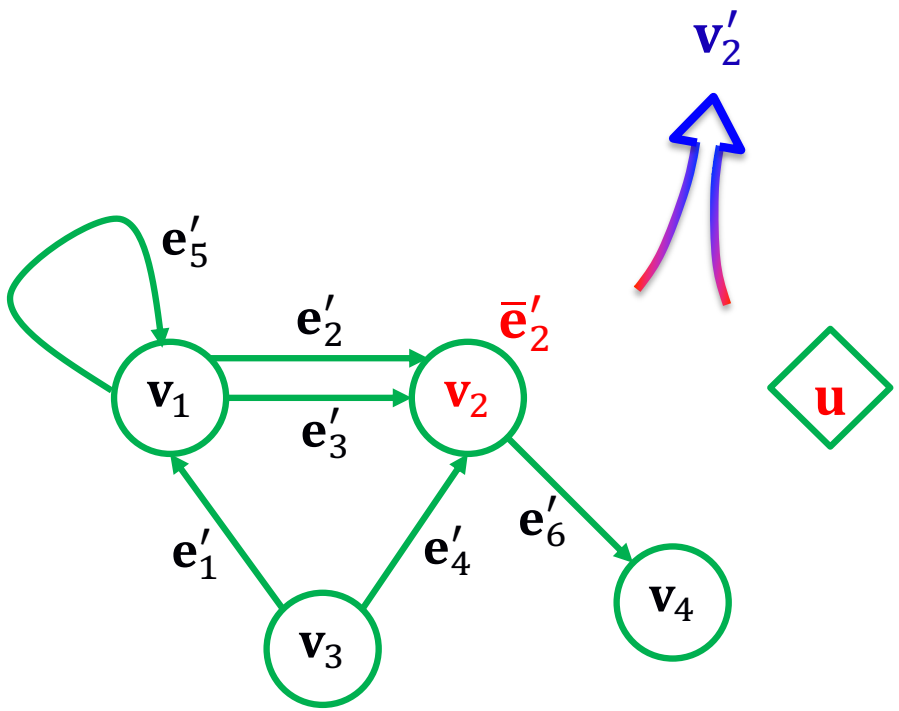
set of incoming edges to node i

$$E'_i = \{(\mathbf{e}'_k, i, s_k)\}_{k=1:N^e}$$

$$\bar{\mathbf{e}}'_i = \rho^{e \to v}(E'_i)$$

aggregation function

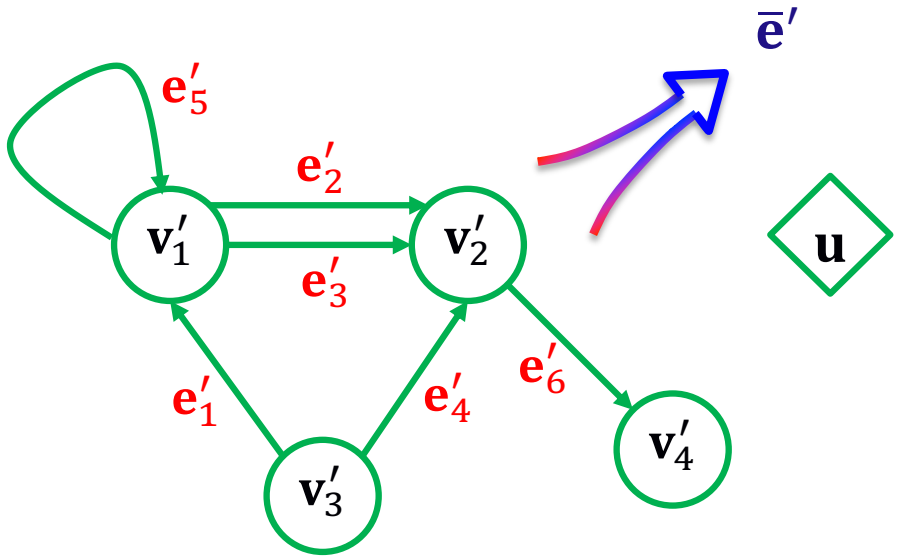
Node update



Apply to all node attributes (can be done in parallel)

$$\mathbf{v}_i' = \varphi^v(\bar{\mathbf{e}}_i', \mathbf{v}_i, \mathbf{u})$$

Global aggregation of edge updates



Aggregate information across all updated edges.

Aggregation function must be invariant to permutations of the inputs.

Typically use sum, mean, or maximum. For example, using maximum:

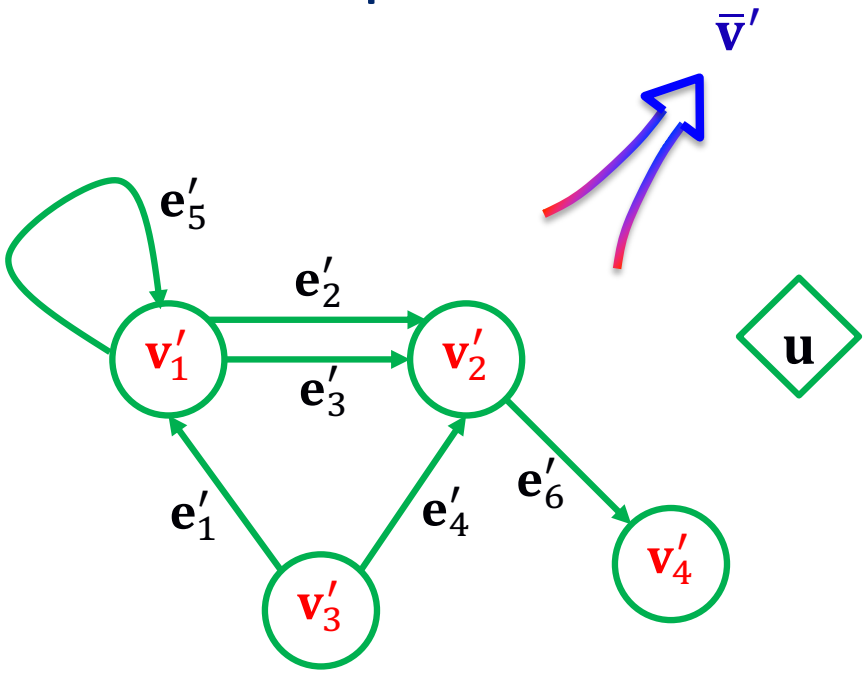
$$\bar{\mathbf{e}}' = \max(\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3, \mathbf{e}'_4, \mathbf{e}'_5, \mathbf{e}'_6)$$

element-wise maximum

$$E' = \{(\mathbf{e}'_k, r_k, s_k)\}_{k=1:N^e}$$

$$\bar{\mathbf{e}}' = \rho^{e \to u}(E')$$

Global aggregation of node updates



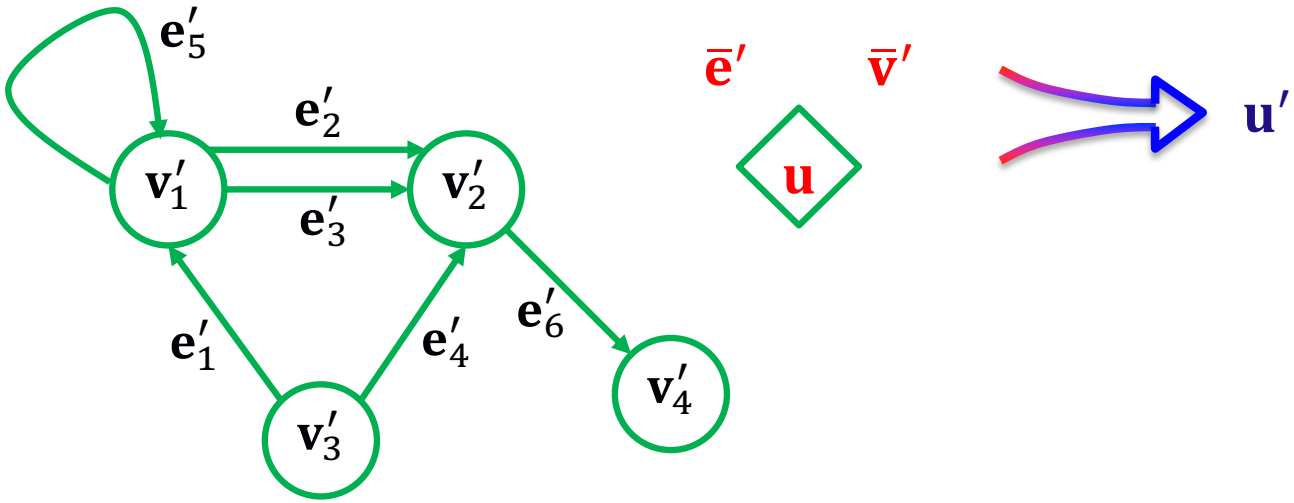
The aggregation function must be invariant to permutations of the inputs. Typically use sum, mean, or maximum. For example, using sum:

$$\bar{\mathbf{v}}' = \mathbf{v}'_1 + \mathbf{v}'_2 + \mathbf{v}'_3 + \mathbf{v}'_4$$

$$V' = \{\mathbf{v}'_i\}_{i=1:N^v}$$

$$\bar{\mathbf{v}}' = \rho^{v \to u}(V')$$

Global update

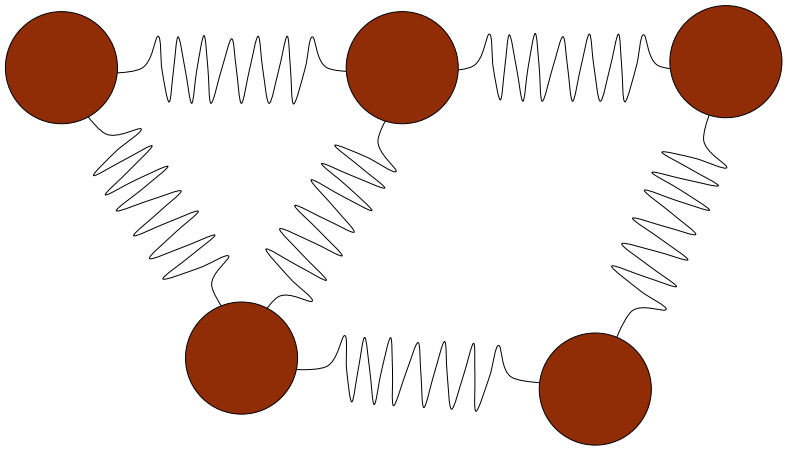


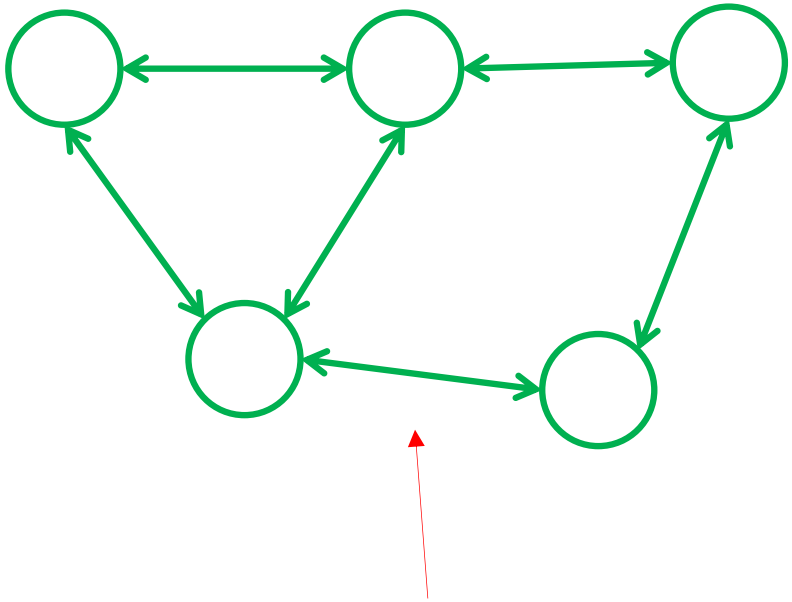
$$\mathbf{u}' = \varphi^u(\bar{\mathbf{e}}', \bar{\mathbf{v}}', \mathbf{u})$$

Example

Node attribute: 3D position & velocity

Edge attribute: force





Rubber balls connected by springs and subject to gravity

Arrows in both directions

Example continued

Edge update	$\mathbf{e}'_k = \varphi^e\left(\mathbf{e}_k, \mathbf{v}_{r_k}, \mathbf{v}_{s_k}, \mathbf{u}\right)$	*Forces between balls*
Local aggregation of edge attributes	$\bar{\mathbf{e}}'_i = \rho^{e\to v}(E'_i)$	*Sum of forces (net force) on each ball*
Node update	$\mathbf{v}'_i = \varphi^v(\bar{\mathbf{e}}'_i, \mathbf{v}_i, \mathbf{u})$	*Position, velocity and kinetic energy of each ball*
Global aggregation of edge updates	$\bar{\mathbf{e}}' = \rho^{e\to u}(E')$	*Sum of per-node net forces (should be zero) and potential energy stored in springs*
Global aggregation of node updates	$\bar{\mathbf{v}}' = \rho^{v\to u}(V')$	*Total kinetic energy of system*
Global update	$\mathbf{u}' = \varphi^u(\bar{\mathbf{e}}', \bar{\mathbf{v}}', \mathbf{u})$	*Net forces and total energy of system*

The functions φ^e, φ^v, φ^u

The three functions at the heart of a GNN can have many forms, such as a single layer neural network, multi-layer (deep) network, and CNN (for image attributes).

Compositions of GN blocks

The node and edge updates in a GN block aggregate data from neighbouring edges/nodes and global attribute.

The ‘receptive field’ is small (ignoring global attribute).

As in a CNN, we increase the scale by chaining layers of GN blocks.

$$G_0 \longrightarrow \boxed{GN_1} \longrightarrow G_1 \longrightarrow \boxed{GN_2} \longrightarrow \cdots \longrightarrow \boxed{GN_M} \longrightarrow G_M$$